



Education

Present 09.2023	Master in Energy Science and Technology, École Polytechnique Fédérale de Lausanne Section of Electrical and Electronical Engineering GPA: 5.5/6.0
06.2023 09.2019	Bachelor in Energy Storage Science and Engineering, Xi'an Jiaotong University Faculty of Electronic and Information Engineering GPA: 91.16/100

Work Experience

09.2024 07.2024	Summer Internship in Charging System (100%) @ SAIC Motor - Performed state-of-the-art review of on-board chargers, conducted a feasibility study on the vertical integration of on-board chargers. - Designed control systems and simulation models for a 6.6 kW/400 V totem-pole bridge-less PFC converter and LLC resonant converter demo.
09.2022 03.2022	Internship in Energy Storage Market (40%) @ China Energy Storage Alliance (CNESA) - Maintained and updated CNESA's energy storage project, policy and bidding information databases . - Collected energy storage news articles; edited drafts of research reports and compiled quarterly global energy storage market tracking report.

Sills & Languages

Relevant Skills	Convex optimization, Power system analysis (pandapower, PyPSA), Smart grid technologies (monitoring, forecasting, optimal power flow) , Energy system modelling
Other Skills	Control design (PID and model predictive control), Data analysis, Git for version control, L ^A T _E X, Life cycle assessment, LTSpice, MS Office, PLECS, System identification
Coding	Python, MATLAB & Simulink, C/C++, LabVIEW
Language	Chinese — Native English — C1 Français — A2

Projects

- SP at DESL: A Framework for Carbon Flow Tracing in Swiss Power Distribution Grids (Ongoing): Upgraded a primitive framework for carbon footprint tracing from inferred Swiss low-voltage to medium-voltage grids. Expanded the capabilities of the framework by upgrading naive generation models by integrating geographical and meteorological information data on roof suitability (for PV modelling).
- SP at DESL: Li-Ion Cell Equivalent Circuit Model Parameters Estimation from Time-Domain Measurements: Utilized voltage and current signals sampled by battery management system for estimating the parameters of battery equivalent circuit models. Explored the identifiability of different parameters and developed a low-frequency model for estimating constant phase element via grey-box estimation.
- The Impact of COVID-19 on Green Energy Transition in European Countries: Delved into the interplay between the COVID-19 and renewable transition in 6 EU countries, analyzed datasets on generation, interests, mobility and government intervention, uncovered how the pandemic impacted electricity generation, consumption, and awareness of renewable energy.
- Cooling System Design of a PV Charging Demo: Conceptualised and developed a PV charging demo with thermoelectric coolers for heat dissipation, achieved real-time monitoring of PV generation and battery SoC, stabilised panel temperature fluctuations within 1°C and increased efficiency by 23%.